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Producing the Ultimate from the Saih Rawl Shuaiba Reservoir

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Abstract

Horizontal drilling of multilateral wells within the last five years has rapidly developed the Saih Rawl Shuaiba reservoir. These wells consist of up to 7 branches, and expose more than 10 km openhole of reservoir per well. Current production of the reservoir is around 8,000 m³/d of oil with an average watercut of 75%. A “produce-the-ultimate”-initiative was followed to systematically analyze, debottleneck and optimize every detail of the production chain: from water injection manifold, water injectors, to oil producers with ESP, including the waterflood efficiency. This has resulted in a range of optimization activities such as ESP changeout, flowline looping, water injector tubing resizing, automation of chokes and an improved water injection distribution system. Initial gains from part of these activities amount to more than 1,000 m³ of oil per day within the first six months.

Introduction

The Saih Rawl oilfield situated in Central Oman, was discovered in 1971 by SR-2 and came on stream in 1975 (see figure 1). The field is oil-bearing in Shuaiba, Mafrq, Gharif and Al Khlata reservoirs, while gas and condensate are produced from the deeper Amin and Barik intervals.

Production started from the most productive Gharif reservoirs. Attempts to produce the Shuaiba reservoir with vertical wells were made from 1984 onwards with three drainholes (SR-16, 20 and 22). However, the tight nature of the reservoir and the thin oil column underlain by water have jeopardized this approach, due to early water production and low initial rates. The development using the conventional vertical well design of that time was not economical.

The current Shuaiba reservoir development has started with bare-foot horizontal multilateral wells from mid-1993 onwards. Due to very limited aquifer support, water injection was implemented from end-1994 with horizontal, multilateral barefoot injection wells located below the OWC. To date 50 production and 20 injection wells have been drilled at lateral spacing from initially 250 m to now 60 m. This adds up to more than 172 km of producing and 104 km of injecting drainhole.

The current typical well design is a 4 to 7-legged multilateral well with a cased 7”-backbone and 6 1/8” openhole legs. Production wells are completed with ESP and deliver between 1,000 to 3,000 m³/d gross. Apart from drilling new wells, development activities have included side-tracking single horizontal to multilateral producers, restimulating existing wells, isolating a water-cycling leg (in injector well) and converting gas-lifted wells to ESP.

The current oil production is around 8,000 m³/d, for a total gross rate of 35,000 m³/d; this represents a field water-cut of 75%. Pressure support is performed at voidage replacement and the reservoir pressure is currently balanced at around 9,000 - 10,000 kPa. Today almost 11 Million m³ of oil have been produced. Currently 38.5 Million m³ are booked as ultimate recovery out of a STOIP of around 90 Million m³.

All Saih Rawl Shuaiba production is routed to the Saih Rawl production station, where it is commingled with gross production from a number of other reservoirs and fields prior to dehydration.

The net oil is exported via the Qarn Alam Main Pumping station into the Main Oil Line to the terminal at the coast. The separated gas is compressed; some is recycled as gaslift, the balance being exported. All produced water is currently re-injected into the Shuaiba reservoir, together with additional water from local water supply wells, as required. The separation and dehydration facilities at Saih Rawl Production Station are going to be expanded during the third development phase in Q3-2001. This will result in a “no constraints” facility.

Geology and Reservoir

The Shuaiba reservoir is a low relief anticline elongated roughly N-S. It is truncated on its southern side by a NW-SE trending fault with a maximum vertical throw of 30 m.

The Shuaiba formation is 60 m thick with a maximum oil column thickness of 35 m and has Nahr Umr shale as top seal. The dominant lithology in the formation is a porous wackestone (limestone composed of fine-grained matrix and < 30% coarse grains), typically with 27% porosity and 2 mD permeability.

There is a clear distinction between the top 12 m of the reservoir and the bottom 46 m in terms of reservoir quality. The top 12 m show better reservoir qualities (up to 30% porosity and > 10 mD permeability) due to leaching, versus 25% porosity and 1 mD in the bottom part.

The OWC (100% S_w) is at an average depth of 1400 mtv.

Fracturation. Fractures within the reservoir were detected on formation imaging logs acquired from more than a dozen horizontal wells located mostly in the crestal part of the field. The interpretation of the image logs revealed two types of fractures; electrically conductive (fractures which are open to a certain extent) and electrically resistive (fractures which are closed or cemented) with an average spacing of 20 to 25 m. The fractures have no detectable influence. This is based on five years production and injection experience, which have not seen any early water breakthrough from injection to production wells due to an open and conductive fracture network. The production mechanism is dominated by the matrix.

This is important in appreciating the following description of the “producing the ultimate” initiative.

Fluid Properties. The Saih Rawl Shuaiba reservoir contains undersaturated oil with a density of 34° API and a dissolved GOR of 20 std m^3/m^3 . The main reservoir fluid characteristics are shown in table 1 of the appendix.

The moderate oil viscosity contributes to a mobility relatively favourable to oil displacement. The mobility ratio is calculated from viscosity and relative permeability end-point as close to 1.6 at bubble point pressure.

Aquifer Influence. Indications based on reservoir simulation work indicate that there is a weak aquifer influx. The influx volume is calculated to be 10-15% of the daily gross offtake. Based on the weak aquifer support, the field development of the Saih Rawl Shuaiba reservoir without any form of pressure maintenance would result in a recovery factor (RF) of 7% only. Therefore a water injection system is required to replace the stock tank gross production on a daily basis to ensure that no decline in mean reservoir pressure occurs below the current stable conditions.

Field Development

Historic Vertical Well Production Performance. Shuaiba reservoir production started via three vertical wells drilled in

1984, 1986 and 1988. These early wells experienced very low productivity and very fast water breakthrough due to water coning. Consequently, development of this thin reservoir was discontinued.

However, with the advent of horizontal drilling in 1991, a well was sidetracked horizontally to a total length of 350 m. The performance of the horizontal sidetrack was promising, starting with an initial oil production rate of 120 m^3/d ; three times the initial rate of the original vertical wells.

Horizontal Well Drilling. Based on the success of the horizontal sidetrack, a reservoir simulation study was performed, which recommended an initial field development based on horizontal wells drilled at a lateral spacing, the distance between two production wells, of 500 m. Only areas with an oil column thickness greater than 20 m were to be developed.

Drilling of additional horizontal production wells started in 1993 with the drilling of seven wells. Reservoir total production increased steeply from less than 100 m^3/d early 1993 to over 2000 m^3/d by the end of the same year. Despite the growth in production, the wells declined rapidly in gross production, which was a clear indication for the lack of pressure support. Nevertheless horizontal drilling was the answer to early water breakthrough and low productivity experienced in the vertical wells drilled in the 1980s (see figure 2).

Waterflood. A trial horizontal water injector was brought on stream in 1994, which resulted in excellent production response due to increased reservoir pressure. A water injection pilot in 1995/96 followed the trial, drilling additional five dual injection wells and four single production wells. These five successful water injection wells, together with the newly drilled production wells, were the main contribution to the reduction of the production offtake decline. A production / injection well lateral spacing of 125 m was implemented after this drilling phase. In 1998 the spacing was revised to 80 m between injection and production well legs. The 1999/2000 drilling campaign followed this pattern.

Currently Saih Rawl Shuaiba comprises of 47 active horizontal production wells and 19 active horizontal injection wells. Beside the flank, where the oil column is less than 10-15 m, the entire field has been covered by production/injection well pairs (see figure 3).

Reservoir Sweep mechanism. The sweep mechanism for a typical Saih Rawl Shuaiba producer/injector pair is represented in figure 4, which shows a cross-section of the oil saturation simulated with a 2D sector model. The mobility ratio of 1.6 results in a relatively efficient waterflood.

The main sweep mechanism around the producer is from water coning. Oil is swept by water in the cone-shaped volume located directly underneath the producer. The longer the horizontal sections the larger the reserves.

The function of the water injection well is often seen as only providing the pressure support that allows the producer to

maintain its liquid production. However, there is also a water sweep in a semi-cylinder located above the injection well, where oil is displaced by the dynamic forces of the injection. There again the drilled injection well contributes on its own to a significant amount of additional reserves.

From these two mechanisms, it appears that the remaining oil in such a configuration is located at the top of the reservoir, in infill locations between wells, justifying studying infill drilling down to very tight spacing.

Current Well Design & Completion. All Saih Rawl Shuaiba wells are drilled horizontally within the reservoir and since 1997 only multilateral (ML) wells were drilled (see figure 5). These ML wells consist of up to seven legs and access more than 10 km (6 1/8" OH) of the reservoir.

The 13 3/8" surface casing is set in the Natih formation at around 900 mtv and cemented to surface to isolate shallow aquifers and provide a barrier against deeper hydrocarbon bearing formations. The following 9 5/8" production casing is run through the Nahr Umr into the Shuaiba reservoir and landed at 90° inclination at a depth of 1380 mtv either at the top of the reservoir (production wells) or at the initial OWC (injection wells). A 7" cemented liner is run across the Shuaiba and serves as a backbone for the 6 1/8" OH legs which exit from there.

Production wells. Up to seven multilateral legs have been drilled from a single motherbore in a production well. Most of the wells are all equipped with high rate, high horsepower electrical submersible pumps (ESP); a few of the wells are still gaslifted. They pump up to 3,000 m³/d gross production from a single well and consume up to 450 kW power. The pump is run on a 4.5" tubing (see figure 6).

Injection wells. Currently up to 4 laterals were drilled from a water injection well motherbore. Historically these wells were equipped with a 4.5" plasticcoated tubing, but that has changed recently to a 7" monobore completion (see figure 7).

Producing the Ultimate Initiative

Production optimization opportunities are immense in such fields like Saih Rawl Shuaiba. The aim of the "Producing the Ultimate" initiative was to identify a long-term strategy on how to produce the developed reserves at its optimum.

Based on the production history, the reservoir can be described as a homogenous carbonate reservoir without flow contribution from a fracture network. Therefore the production rate defining key parameters were considered to be the productivity and injectivity index, reservoir limitations as bubble point and formation fracture pressure, as well as the existing surface and subsurface infrastructure.

Analyze your production system data. Productivity index and injectivity index are very similar between wells due to the relatively homogenous nature of the reservoir.

Productivity Index. Derived from static pressure surveys and back-calculated bottomhole pressures during production conditions, from gaslifted or ESP-lifted production wells, the

average field specific productivity index (SPI) was determined to be 0.08 m³/d/kPa/kmOH.

Injectivity Index. Water injection wells are drilled below the OWC. The specific injectivity index of the injection wells was derived at around 0.04 m³/d/kPa/kmOH. The tighter limestone matrix at the depth of the water injection wells can explain this.

Surface Infrastructure. All production wells have their own 6" flowline to the manifold at the station. The water injection wells have an individual injection line from the water injection manifold to the wellsite. The typical flowline length for both production and injection wells varies between 1–5 km. The production manifold is operated at around 400 kPa; whereas the water injection pumps can deliver a maximum of 8,000 kPa discharge pressure; with the current throughput of injection water the water injection manifold is operated at around 6,800 kPa.

Identify your boundaries. In order to be able to produce the ultimate from a reservoir the production boundary conditions need to be clearly defined. The technical limit is not necessarily the optimum.

Bubble Point Pressure. Saih Rawl Shuaiba oil is undersaturated. Producing the reservoir below bubble point would result in gas-release into the pore space and thus a build-up of a residual gas saturation. This would have an enormous impact on the production behavior of the horizontal ESP completed production wells. Three-phase flow along undulating horizontal wells would result in heavily slugging wells and ultimately lead to premature ESP failures.

Moreover the reduction of the reservoir pressure below the bubble point would lead to a change in relative permeability and thus a reduction in productivity and ultimate recovery from the field. It was therefore decided to operate the production wells at the bubble point pressure.

Formation Fracture Pressure. The entire production performance of this reservoir is based on a homogenous tight matrix carbonate without a fracture network contributing to flow. Even sweep is the key to produce the field the way it is developed. Exceeding the fracture pressure of a water injection well would result in an open conduit towards the supported production well, whereby the fracture would have a conductivity much larger than the matrix. Injection water recycling up to 100% would be the result. Therefore it is crucial to limit the injection pressure to below the fracturing pressure of the formation.

Define your strategy. Based on the identified boundaries for production and injection wells, the production strategy for Saih Rawl Shuaiba was defined as follows: "Produce at bubble point pressure and inject just below formation fracture pressure" (see figure 8). This results in the highest production rate and developed reserves without jeopardizing the ultimate recovery.

Analysis of reservoir compartments. In order to perform a systematic analysis of the reservoir, well and infrastructure

performance, the Saih Rawl Shuaiba reservoir was subdivided into several small, more or less independent compartments. The dependency between production offtake and water injection was used to define these compartments. These reservoir compartments were individually analyzed with respect to production optimization as well as reserves developed.

Applying the strategy to the Saih Rawl Shuaiba production system. Every production and injection well within the Saih Rawl Shuaiba reservoir was reviewed with respect to the defined boundaries of fracture pressure and bubblepoint pressure. For every well that could not fulfill the strategy requirements, the respective activity to overcome this underperformance issue was economically evaluated.

Maximizing production. ESP's lift 85% of the Saih Rawl Shuaiba production. Therefore the focus was clearly on ESP performance. Potential flowline debottlenecking of high rate production wells was addressed by designing the ESP's for higher backpressure. This can be achieved much quicker and more cost-effective than looping of flowlines. The lift performance of the gaslift production wells was reviewed and production wells, which could not be produced at bubble point pressure, were consequently converted to ESP production wells.

ESP upsize and automated production. The focus on ESP performance and the bubble point pressure criteria led to a series of ESP upsizes to higher production rates. The average ESP for this field can be defined as a 12,000 bbl/d ESP with around 300 kW installed, but 20,000 bbl/d ESP with 450 kW have also been installed. All ESP's were designed against a backpressure of around 2,000 kPa and a watercut of 95%. This resulted into enough head delivered by the ESP to automate the entire production system by installing a choke which is remotely operated. All ESP wells are equipped with remote telecommunication units to allow for full computer assisted operations. The production wells can be started up and shutdown from the office in the field. Monitoring of the ESP performance does not require a site visit anymore, which resulted in reduced HSE exposure of the field personnel.

Maximizing water injection. In order to inject the maximum amount of water into a particular well, it needs to be injected slightly below the fracture pressure of the formation. However the "producing the ultimate"-initiative identified a number of wells that were constrained by the pressure drop in the tubing and/or the flowline. This can be overcome by installing larger tubing or by looping of the existing flowline, as these options are significantly cheaper than changing out the high cost water injection pumps.

Tubing upsize. Some high rate injection wells with a short flowline were found to be injection constraint by the standard 4.5" plastic coated tubing only. A water injection well that requires 3,000 m³/d injection rate to support the production offtake, gains around 2,700 kPa pressure by a tubing upsize to 7". This results in additional water injection of up to 100 m³/d per km openhole and gains on average around 50 to 100m³/d

from the supported production wells. Moreover a 7" tubing completion results in a monobore completion from the wellhead down to TD of each individual leg. Much easier and faster intervention for logging and workover purposes is the outcome.

Flowline looping. In a few cases the flowline needs to be looped or upsized, since the water injection well is far away from the manifold. The pressure drop along the flowline is already so high, that the tubinghead pressure is insufficient to inject the required rate into the reservoir. This kind of activity is a relatively unattractive activity though.

Example SR123/I27. SR123 was drilled in November 1999 as a multilateral well with seven openhole legs prior to the PTU initiative. In total 10,720 m of reservoir were accessed. The well was completed with a 12,000 bbl/d ESP due to the unavailability of a larger pumpsystem (ESP and auxilliary equipment). An initial production rate of 1,100 m³/d at 20% watercut was tested. The supporting water injection well was brought on stream 2.5 months later, completed with a 4.5" tubing size, which brought the heavily depleted reservoir pressure back to field average.

Meanwhile the production well was underperforming, the bottomhole flowing pressure was significantly higher than the bubblepoint pressure and the ESP was operating at its production limit. Immediately after the larger ESP became available the well was worked over and a 20,000 bbl/d ESP was installed as part of this initiative. This was the first ESP installation of this size in PDO ever.

In conjunction with the higher offtake from SR123, the corresponding water injection well SR127 saw increased injection rates due to lower average reservoir pressure in that compartment. The maximum bottomhole injection pressure could not be achieved anymore and debottlenecking of the system was required. The analysis showed that a tubing upsize to a 7" monobore completion would be sufficient to allow for optimal injection rate and pressure.

The ESP and tubing upsize gained 350 m³/d initially and delayed the BSW development (see figure 9).

Remediate sweep inefficiency. The reservoir management of the active waterflood development of Saih Rawl Shuaiba is based on even water injection distribution across the entire field and even production inflow from the horizontal production wells. Improper stimulation "spot" techniques of multilateral OH legs resulted in uneven sweep. Rapid watercut increase and early water breakthrough in production wells has been seen in a few occasions. A number of production and injection logging runs were performed to diagnose potential sweep inefficiencies. Due to the limited production logging options via ESP completed production wells, the identification and treatment of uneven sweep is foreseen via injection wells. Full access and no deferment from these wells is given to identify uneven injectivity and therefore most likely uneven sweep and inflow into the concurrent production well.

SR115 challenge. SR115, a four-legged horizontal water injection well, supports mainly the two ML horizontal producing wells SR108 and SR111. A rapid watercut

development of SR111, compared to SR108, was suggesting an uneven injection water distribution within SR115, most likely via a fracture, induced during an initial step-rate test. This was confirmed by a injection logging run performed on CT (see figure 10)

During a workover in December 1999 the loss zone was identified in leg#2 of SR115 by an innovative application of an existing production logging principle. The logging tool was pumped through the drillpipe and with a constant injection rate into the annulus at surface, the water flow velocity behind the drillpipe along the openhole section could be measured. This logging principle was performed for the first time in Oman and the Middle East (see figure 11). It adds significant value to the business, since it enables us to log the entire length of OH water injection wells, which is beyond CT reach. The workover itself was unsuccessful in shutting of the loss zone with a simple cement plug and therefore production from SR111 could not be regained. SR111 was closed in since January 2000 until a scheduled workover in August 2000 to isolate mechanically leg#2 of SR115 at the window position by a hydraulically set casing patch. As a result SR111 production of around 150-200 m³/d net oil could be regained.

A 7" plastic coated tubing was run, which resulted in the first Monobore completion in the Saih Rawl field and within the Qarn Alam Asset.

This casing patch installation was the first ML window shut off in PDO and proved to be a reliable option for all mechanical shut-off challenges in the matrix carbonate ML wells of the Qarn Alam asset.

Additional logging of water injection wells is envisaged to cure production misbehavior seen by unfavorable watercut developments.

Stimulation and Cleanup Treatments. Historically all horizontal production and injection wells were stimulated with acid. The acid recipe, volumes and the treatment technique though changed with time.

In the early field development phase, the horizontal sections were only up to 1000 m long and relatively small volumes (between 3-5 m³) of 28% HCl were pumped in several stages of around 200 m coverage each.

With the evolution of longer OH sections and multiple legs, the treatment volumes per stage and the number of intervals increased. The specific treatment volume was kept constant of around 20 l/m openhole.

As of end 1995 huge volumes of non-diverted acid (up to 95 m³) were bullheaded into the openhole section after the entire interval had been stimulated already in several stages with smaller volumes. This acid-spot technique together with a final bullheading of acid resulted obviously in very uneven acid coverage of the open hole section. Rapid watercut increases have been monitored in wells treated in this manner, which could not be explained by the reservoir simulation model. To determine this unusual behavior, a PLT was run on a CT in SR68 in December 1998. The result showed clearly two waterinflux zones, both correlating to places where acid had been pumped. Moreover the PLT-caliper showed an

increased wellbore diameter at the position where 95 m³ of acid was bullheaded after the initial spot acidization, compared to the caliper taken during the drilling phase.

These results highlighted the need for a different approach in stimulation and clean-up of production and injection wells. The main objective was to achieve an even clean-up of the wellbore to allow for even production or injection along the long horizontal intervals. Due to the fact that the permeability decreases with depth, the requirement of stimulating water injection wells was clear.

On the other hand the need for stimulating production wells was questioned and field trialled in a new dual-lateral well, sidetracked from an existing single horizontal production well. One leg of this well was acidized with a retarded acid-in-oil emulsion with pH-dependent diverting acid agent. The other leg was cleaned up with an oxidizing emulsion, often referred to as bleach; in this case Calciumhypochlorid was used. This oxidizer acts on the organic material within the filtercake and breaks the polymer chains down first; the remaining oxidizer will attack the starch in the filtercake. The results were eye opening. The productivity index, and therefore the gross production, of the acidized leg was twice as high as the bleached leg, a clear indication for an effective matrix treatment. But both legs produced the same amount of net oil. The acidized leg never produced at a watercut lower than 55%, compared to 25% watercut for the bleached leg. In addition the watercut increase of the acidized leg was worse. So, by bleaching production wells, not only the same net oil rate, but also more even sweep and therefore a higher recovery factor was achieved. As a result of these findings, all subsequent production wells were cleaned up after drilling with Calciumhypochlorid. No more unexpected rapid watercut increases have been witnessed since then.

Water injection wells still require matrix stimulation. This was proven by one unsuccessful bleach attempt in a multilateral water injection well that resulted in an extremely low injectivity index for that well. It was realized that a simple filtercake removal treatment was insufficient to guarantee proper pressure support for the concurrent production wells. A restimulation of this well was required.

Therefore the acidization technique and recipe for the following water injection wells were changed back to a retarded acid-in-oil emulsion system. The data gathered during the placement of the acid system indicates far better zonal coverage than with the previous systems.

Future Field Development

Future plans for the Saih Rawl Shuaiba reservoir are to increase the injection capacity to 50,000 m³/d in 2001 and to 75,000 m³/d the years thereafter while increasing the gross offtake to the same amount, to drill additional infill production and injection wells to increase recovery of attic oil and to target oil volumes contained in the flanks of the reservoir.

Automation of the production system. To produce and inject at all times close to the limits of formation fracture and bubble point pressure an automated intelligent system is

required. The automated chokes on the production wells are controlled by the pump intake pressure of the ESP, which is transmitted via the ESP cable from the downhole pressure gauge to the control device on surface. Changes of the pump intake pressure and therefore the bottomhole flowing pressure are reflected in automated choke manipulations to allow for continuous optimal production at the lower limit; the bubble point pressure at reservoir depth.

On the water injection side a manual system is currently in place. The operators are requested to manually adjust the choke against a certain bottomhole injection pressure, which is derived from pressure drop calculations along the flowline and the tubing and is calculated based on actual flow and pressure measured at the water injection manifold.

Currently a modeling software of the Saih Rawl Shuaiba production system is developed for automated operations and continuous optimization based on actual well performance and the dependence between production and injection wells in the field. Together with the currently ongoing installation of actuators on the existing chokes a fully automated water injection system is envisaged for the first half of 2001.

Ringfencing a technology trial area of the field. Ringfencing a part of the existing developed matrix carbonate reservoir is required to trial a number of further development options for the future of Saih Rawl Shuaiba. This compartment was identified in a relatively under-developed crestal part of the reservoir.

Currently ongoing activities are infill drilling to 40m spacing, with a 1:1 producer-injector ratio. However, other activities are foreseen such as closing in high watercut wells, or converting production wells to water injection wells. Less conventional methods such as coiled-tubing drilling of fishbones from existing production legs, or EOR methods such as surfactant flooding could contribute to a better sweep efficiency. These development options will to be evaluated, not only in a predictive manner (using reservoir simulation), but also through a field trial in this reservoir compartment.

Moreover an intense data-gathering program is planned in this trial compartment to maximize the understanding of the pilot processes. This will include coring, saturation logging, production logging, and PVT sampling. Particular production monitoring needs might be identified in the course of this trial.

Drilling Opportunities. Infill drilling opportunities have been identified from analysis of the waterflood efficiency in the field. The analysis was done for each individual reservoir compartment. Drilling additional production or injection wells is based on the current injection/offtake relation and the overall developed reserves for each compartment. For 2001 both infill production and injection wells are planned, expected to gain approximately 2,500 m³/d net oil. For this development phase, it is planned to drill infill production and injection legs to a 40m spacing. Additional injection legs are also planned in areas of the field where a water injection deficit was identified.

Conclusions

A systematic analysis of an entire production system will result in substantial production gains, like the more than 1000 m³/d initial gains after six months in the Saih Rawl Shuaiba case. It results a detailed quantitative information about bottlenecks that can subsequently be used to justify optimization activities. Application of innovative technologies for production logging, stimulation treatments, multilateral completions as well as the automation of the production system, are enablers for achieving ultimate production performance from a reservoir.

Acknowledgements

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Nomenclature

<i>BSW</i>	= Bulk Solids & Water
<i>CT</i>	= Coiled Tubing
<i>DP</i>	= Drill Pipe
<i>E</i>	= East
<i>EOR</i>	= Enhanced Oil Recovery
<i>ESP</i>	= Electrical Submersible Pump
<i>GOR</i>	= Gas Oil Ratio
<i>HCl</i>	= Hydrochloric acid
<i>J</i>	= Productivity Index [stock-tank m ³ /(d·kPa)]
<i>LWD</i>	= Logging While Drilling
<i>mtv</i>	= Meters true vertical
<i>ML</i>	= Multilateral
<i>N</i>	= North
<i>OH</i>	= Open Hole
<i>OWC</i>	= Oil Water Contact
<i>PDO</i>	= Petroleum Development Oman
<i>PLT</i>	= Production Logging Tool
<i>PTU</i>	= Producing the Ultimate
<i>PVT</i>	= Pressure Volume Temperature
<i>RF</i>	= Recovery Factor
<i>S</i>	= South
<i>S_w</i>	= Saturation, water
<i>SPI</i>	= Specific Productivity Index
<i>STOIIP</i>	= Stock Tank Oil Initially In Place
<i>SR</i>	= Saih Rawl field in Central Oman
<i>TD</i>	= Total depth
<i>W</i>	= West
<i>WSW</i>	= Water Supply Well

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SI Metric Conversion Factors

°API	141.5/(131.5+°API)	=	g/cm ³
bbl	x 1.589 873	E-01	= m ³
cp	x 1.0	E-03	= Pa·s
mD	x 9.869 23	E-04	= μm ²
ft	x 3.048	E-01	= m
ft ³	x 2.831 685	E-02	= m ³
°F	(°F-32)/1.8	=	°C
in	x 2.54	E-00	= cm
psi	x 6.894 757	E-00	= kPa
scf/bbl	x 1.801 175	E-01	= std m ³ /m ³
HP	x 0.7457	E-00	= kW

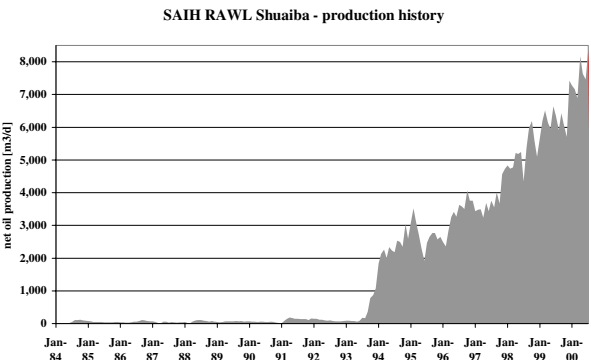


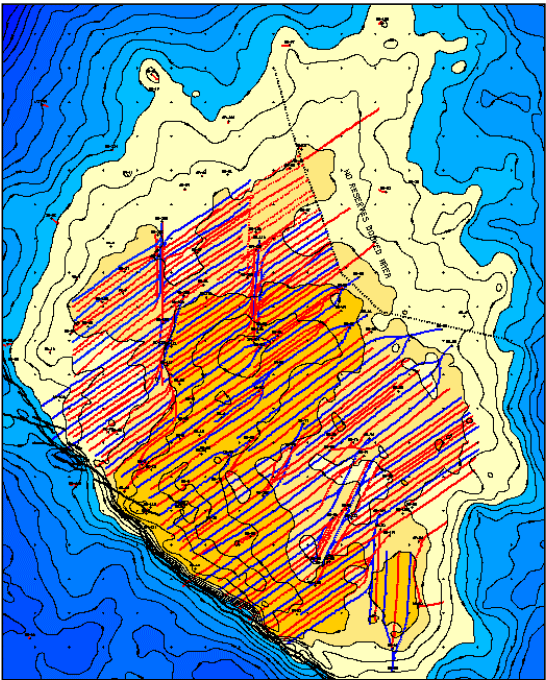
Fig. 2 – Saih Rawl Shuaiba production development 1984-2000

Appendix

Tables

TABLE 1	Saih Rawl Shuaiba reservoir data
Reservoir depth	1380 mtv
Reservoir temperature	74 °C
Initial reservoir pressure	15,170 kPa
Bubble point pressure	5,171 kPa
Oil density	34° API
Dissolved GOR	20 std m ³ /m ³
Formation volume factor	1.1
Oil viscosity	1.53 mPa.s
Water viscosity	0.45 mPa.s
Mobility ratio	1.6

Fig. 3 – Saih Rawl Shuaiba: Top structure map illustrating horizontal multilateral production and injection wells



Figures

Fig. 1 – Saih Rawl field location in the Sultanate of Oman

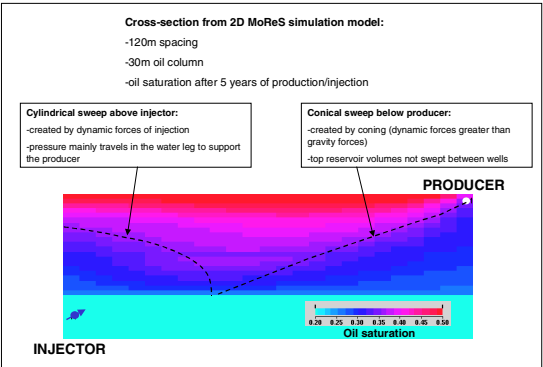
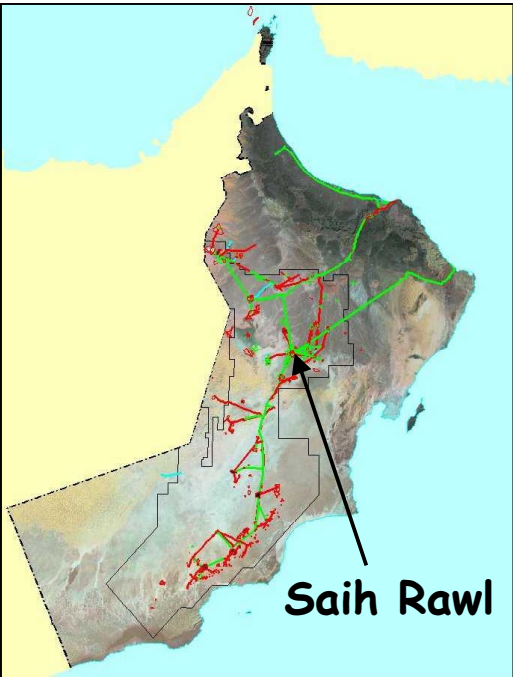


Fig. 4 – Reservoir sweep model

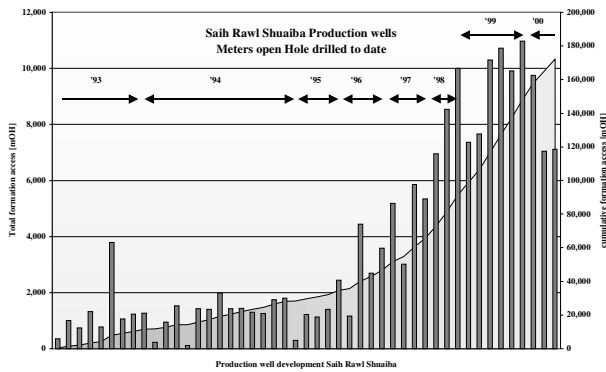


Fig. 5 – Evolution of Saih Rawl Shuaiba horizontal multilateral drilling

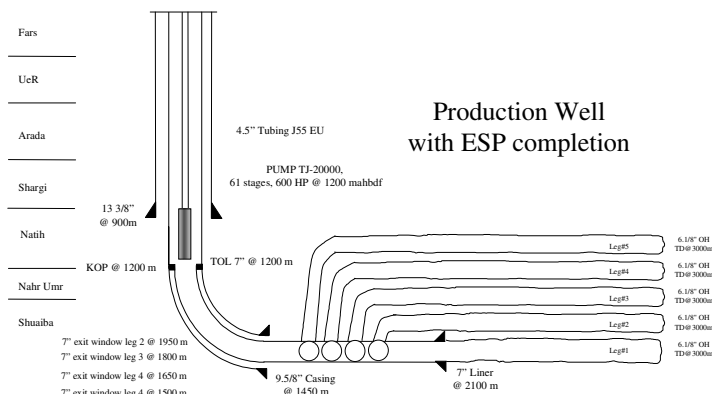


Fig. 6 – Typical Saih Rawl Shuaiba production well design (e.g. 5-leg multilateral well with ESP completion)

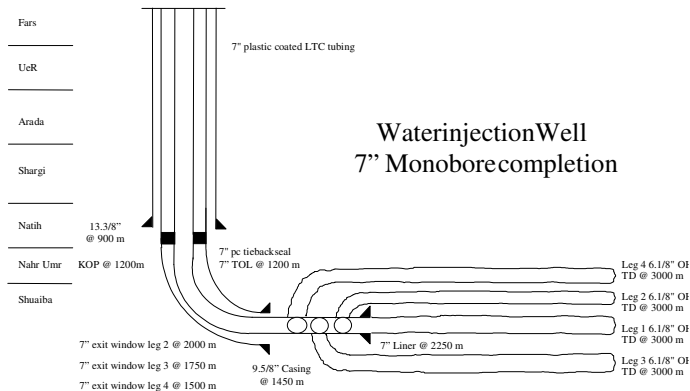


Fig. 7 – Typical Saih Rawl Shuaiba injection well design (7-inch Monobore completion)

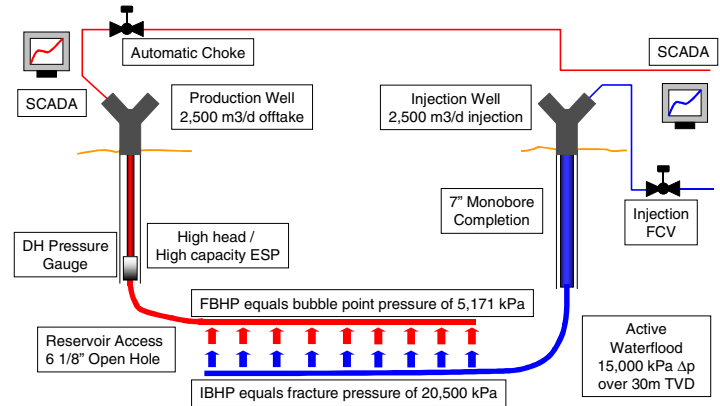


Fig. 8 – Producing the Ultimate™ concept

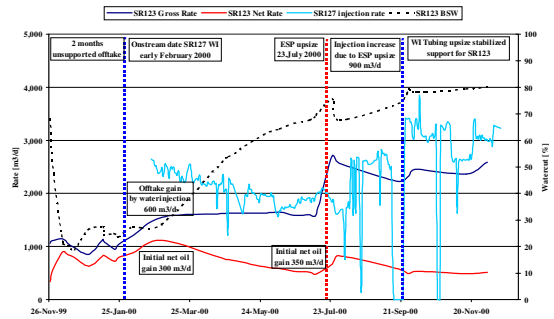


Fig. 9 – Producing the Ultimate™ example: SR123/SR127 reservoir section

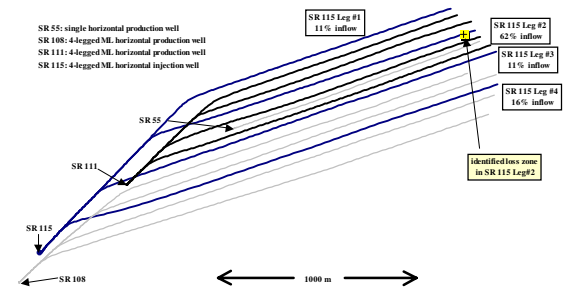


Fig. 10 – SR115 challenge: Identification of a loss zone in a four-leg multilateral water injection well

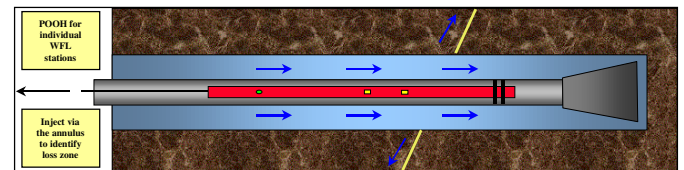


Fig. 11 – "Pump-through-drillpipe" injection logging concept